

STT 861: Theory of Probability & Statistics I

Lecture 8: Density & Mass Functions

Guanqun Cao

Department of Statistics & Probability
Michigan State University

Sep 24, 2026 — Wells Hall C506

Reference: Casella & Berger Ch. 1.6

Lecture 8 Outline

Today

- Probability mass function (PMF) – discrete case
- Probability density function (PDF) – continuous case
- Relation between CDF and PMF/PDF
- Validity conditions

Definition 1.6.1 — Probability Mass Function

Definition 1.6.1 (CB): PMF

The **probability mass function** (pmf) of a *discrete* random variable X is the function

$$f_X(x) = P(X = x) \quad \text{for all } x.$$

PMF and CDF relationship

The pmf gives us *point probabilities*. We recover the cdf by summing:

$$P(X \leq b) = F_X(b) = \sum_{k=-\infty}^b f_X(k) \quad (\text{discrete case})$$

In particular, for any positive integers $a \leq b$:

$$P(a \leq X \leq b) = \sum_{k=a}^b f_X(k)$$

The **size of the jump** in the cdf at point x equals $f_X(x) = P(X = x)$.

Notation convention

Uppercase letter for the cdf (F_X), lowercase for the pmf or pdf (f_X). We write $X \sim f_X(x)$ or $X \sim F_X(x)$ to mean “ X has this distribution.”

Example 1.6.2 — Geometric PMF

Example 1.6.2 (CB): Geometric probabilities

From Example 1.5.4 (toss a coin until a head): X = number of tosses, $P(\text{head}) = p$.

PMF:

$$f_X(x) = P(X = x) = (1 - p)^{x-1} p, \quad x = 1, 2, 3, \dots$$

Recovering probabilities from the PMF: For integers $a \leq b$:

$$P(a \leq X \leq b) = \sum_{k=a}^b f_X(k) = \sum_{k=a}^b (1 - p)^{k-1} p = F_X(b) - F_X(a - 1)$$

Key point

The pmf $f_X(x) = P(X = x)$ is the size of the jump in the cdf at x . Summing the pmf gives the cdf. The pmf and cdf carry **identical information**.

From Discrete to Continuous: Why Sums Become Integrals

Why $P(X = x) = 0$ for continuous X

For a *continuous* F_X , for any $\varepsilon > 0$:

$$P(X = x) \leq P(x - \varepsilon < X \leq x) = F_X(x) - F_X(x - \varepsilon)$$

Taking $\varepsilon \downarrow 0$ and using continuity of F_X :

$$0 \leq P(X = x) \leq \lim_{\varepsilon \downarrow 0} [F_X(x) - F_X(x - \varepsilon)] = 0$$

So $P(X = x) = 0$ for every x ! We cannot use point probabilities.

The analogy: replace sums by integrals.

	Discrete	Continuous
Point probability	$f_X(x) = P(X = x)$	$f_X(x)$ (density, not a probability)
CDF from density	$F_X(x) = \sum_{k \leq x} f_X(k)$	$F_X(x) = \int_{-\infty}^x f_X(t) dt$
Interval prob.	$\sum_{k=a}^b f_X(k)$	$\int_a^b f_X(t) dt$

Definition 1.6.3 — Probability Density Function

Definition 1.6.3 (CB): PDF

The **probability density function** (pdf) $f_X(x)$ of a *continuous* random variable X is the function satisfying

$$F_X(x) = \int_{-\infty}^x f_X(t) dt \quad \text{for all } x. \quad (1.6.3)$$

Equivalently, by the Fundamental Theorem of Calculus:

$$\frac{d}{dx} F_X(x) = f_X(x) \quad (1.6.2)$$

whenever F_X is differentiable at x .

Important: $f_X(x)$ is NOT a probability

- $f_X(x)$ can be greater than 1.
- $P(X = x) = 0$ for every specific value x .
- **Areas** under f_X give probabilities: $P(a \leq X \leq b) = \int_a^b f_X(x) dx$.
- Since X is continuous: $P(a < X < b) = P(a \leq X \leq b)$.

Example 1.6.4 — Logistic PDF (Figure 1.6.1)

Example 1.6.4 (CB): Logistic

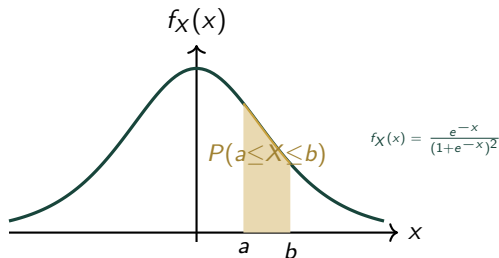
From Example 1.5.5: $F_X(x) = \frac{1}{1 + e^{-x}}$.

PDF by differentiation:

$$f_X(x) = \frac{d}{dx} F_X(x) = \frac{e^{-x}}{(1 + e^{-x})^2}$$

Interval probability:

$$\begin{aligned} P(a < X < b) &= \int_a^b f_X(x) dx \\ &= F_X(b) - F_X(a) \\ &= \frac{1}{1 + e^{-b}} - \frac{1}{1 + e^{-a}} \end{aligned}$$



Theorem 1.6.5 — Characterising Valid PDFs and PMFs

Theorem 1.6.5 (CB)

A function $f_X(x)$ is a pdf (or pmf) of a random variable X **if and only if**:

Ⓐ $f_X(x) \geq 0$ for all x .

Ⓑ $\sum_x f_X(x) = 1$ (pmf) or $\int_{-\infty}^{\infty} f_X(x) dx = 1$ (pdf).

Constructing a PDF from any nonneg. function

Let $h(x) \geq 0$ on a set A , zero elsewhere, with $\int_{\{x \in A\}} h(x) dx = K < \infty$.

Then $f_X(x) = \frac{h(x)}{K}$ is a valid pdf.

Example: $h(x) = x^2$ on $A = [0, 1]$. $K = \int_0^1 x^2 dx = \frac{1}{3}$. So $f_X(x) = 3x^2$ on $[0, 1]$ is a valid pdf.

Section 1.6 — Summary

	PMF (discrete)	PDF (continuous)
Definition	$f_X(x) = P(X = x)$	$F_X(x) = \int_{-\infty}^x f_X(t) dt$
Is a probability?	Yes: $f_X(x) \in [0, 1]$	No: $f_X(x)$ can exceed 1
Point prob.	$P(X = x) = f_X(x) > 0$ possible	$P(X = x) = 0$ always
Interval prob.	$\sum_{k=a}^b f_X(k)$	$\int_a^b f_X(x) dx$
CDF recovery	$F_X(x) = \sum_{k \leq x} f_X(k)$	$F_X(x) = \int_{-\infty}^x f_X(t) dt$
CB example	Ex 1.6.2 (geometric)	Ex 1.6.4 (logistic)

Key Results of §1.6

- ① **Def. 1.6.1:** PMF $f_X(x) = P(X = x)$ for discrete X .
- ② **Def. 1.6.3:** PDF defined by $F_X(x) = \int_{-\infty}^x f_X(t) dt$; equivalently $f_X = F'_X$.
- ③ **Thm 1.6.5:** f_X is a valid pdf/pmf $\iff f_X \geq 0$ and integrates (or sums) to 1.
- ④ Both pmf/pdf contain the same information as the cdf.

Activity 8 – PMF and PDF

Activity 8

- Ⓐ Let X have PMF $f(x) = cx^2$ for $x = -1, 0, 1, 2$.
 - Ⓐ Find c .
 - Ⓑ Find $F(1.5)$.
- Ⓑ Let $f(x) = 3x^2$ for $0 \leq x \leq 1$ (0 otherwise).
 - Ⓐ Verify it's a valid PDF.
 - Ⓑ Find $P(X > 0.5)$.
 - Ⓒ Find the CDF.
- Ⓒ **Bonus:** $f(x) = c/(1 + x^2)$ for all $x \in \mathbb{R}$ (Cauchy). Find c .

Activity 8 – Solutions

- (a) $\sum_{x \in \{-1, 0, 1, 2\}} cx^2 = c(1 + 0 + 1 + 4) = 6c = 1$, so $c = \mathbf{1/6}$.
 $F(1.5) = f(-1) + f(0) + f(1) = 1/6 + 0 + 1/6 = \mathbf{1/3}$.
- (b) (i) $3x^2 \geq 0$ on $[0, 1]$ ✓; $\int_0^1 3x^2 dx = 1$ ✓.
(ii) $P(X > 0.5) = \int_{0.5}^1 3x^2 dx = [x^3]_{0.5}^1 = 1 - 0.125 = \mathbf{0.875}$.
(iii) $F(x) = x^3$ for $0 \leq x \leq 1$; $F(x) = 0$ for $x < 0$; $F(x) = 1$ for $x > 1$.
- (c) $c \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = c\pi = 1$, so $c = \mathbf{1/\pi}$.